

Help Desk Homelab Starter Checklist

Build a lab that teaches support skills instead of collecting noisy hardware trophies. Work top to bottom. Do not buy stage N+1 until stage N boots and stays up for a week.

01. Use the computer you already own until it blocks the lab

- ☐ Confirm the existing laptop or desktop can boot to BIOS/UEFI and install updates.
- ☐ Enable virtualization (Intel VT-x / AMD-V) in firmware for later Docker / VM work.
- ☐ Free 40 GB of disk and confirm it can run one small VM before buying anything.

Buying hardware before you've used what you own is how homelabs turn into shelves of dust.

02. Add storage and adapters that make recovery practice safe

- ☐ One external drive (WD Elements 2TB+) reserved for lab backups and recovery drills.
- ☐ SATA-to-USB adapter cable to read old drives and practice pulling files off dead machines.
- ☐ Two USB flash drives (32 GB+) for bootable installers (Ventoy on one, imaging tool on the other).

One external drive is not a backup plan on its own — it's the first copy. Add a second target before you rely on it.

03. Wire the network before buying more compute

- ☐ TP-Link 8-port unmanaged gigabit switch under the desk.
- ☐ Two short (0.5-1m) Cat6 patch cables for the switch <-> router run and the primary lab host.
- ☐ Label both ends of every cable. Sharpie on a strip of masking tape is enough.

Wi-Fi will make you doubt everything else. Wire the lab so failures point at the lab, not the router.

04. Add a mini PC only when you need a dedicated always-on host

- ☐ N100 or Ryzen mini PC with 16 GB RAM and a 512 GB NVMe.
- ☐ Install Proxmox, Debian, or Ubuntu Server — pick one and stick with it for 30 days.
- ☐ Set up SSH key auth and disable password login on day one.

Buy for idle wattage and quiet fans before benchmark scores. Homelab machines run 24/7.

05. Add a UPS when storage, DNS, or self-hosted services matter

- [] CyberPower CP1500PFCLCD (or equivalent pure sine-wave 1500VA).
 - [] Plug in: mini PC, switch, external drive dock. Nothing else.
 - [] Install NUT or the vendor daemon and test one automatic shutdown before you need it.
- A UPS is insurance, not a battery bank. If you don't have anything worth protecting yet, skip it.*
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Skill drills once the base lab is up

- [] Image a laptop drive to the external drive, wipe the laptop, and restore. Time it.
 - [] Break DNS on a VM (bad /etc/resolv.conf), then diagnose with dig, nslookup, and ping in that order.
 - [] Set a static IP on the mini PC, confirm it survives a reboot, then document your subnet on paper.
 - [] Snapshot a VM, break something on purpose (delete /etc/nginx), restore from snapshot.
 - [] Write a one-page ticket for a fake user issue you just caused. Include repro steps and resolution.
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Do not buy these yet

- A rack. Racks come after you have three things worth racking.
 - A managed switch. Unmanaged is enough until you actually need VLANs.
 - A NAS appliance. Learn on a mini PC + external drive first.
 - 10GbE anything. Your gigabit uplink is fine for a starter lab.
 - RGB fans. They are not the reason the lab isn't working.
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